PRAJWAL MOHAN

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SUMMARY

Biomedical Engineering graduate specializing in medical device development, verification & validation, and biosignal-driven system design. Experienced in wearable medical technologies, clinical AI systems, and healthcare innovation initiatives within academic medical environments. Skilled in test development, quality documentation, and data-driven problem solving aligned with regulated medical device lifecycles. Seeking opportunities in Biomedical R&D, Quality Engineering, Validation, or Systems Engineering within the medical technology industry.

EDUCATION

M.S, Biomedical Engineering Dec 2025
Arizona State University, Tempe, AZ 3.87 GPA

B.S, Biotechnology May 2022
RV College Of Engineering, Bangalore, India 7.62 CGPA

TECHNICAL SKILLS

Biomedical Engineering & Systems: Medical Device Development Lifecycle, Design Verification & Validation (V&V), Biosignal Processing (EEG), Wearable Medical Systems, Closed-Loop Feedback Systems, Human-Centered Device Design, Experimental Testing & Data Analysis
Quality & Regulatory: Quality Management Systems (QMS), Good Documentation Practices (GDP), Root Cause Analysis (RCA), FDA Medical Device Regulatory Concepts, Risk-Based Engineering (ISO 14971 concepts), Test Protocol Development
Programming & Data Analysis: Python (Pandas, NumPy, Scikit-Learn), MATLAB (Signal & Image Processing), Statistical Analysis & Modeling
Engineering Tools: SOLIDWORKS, Tableau, Microsoft Office Suite, Git/GitHub

PROFESSIONAL EXPERIENCE

Research Aide – Strategic Initiatives (ASU Health), Tempe, AZ Jan 2025 – Dec 2025

- Supported development and coordination of large-scale healthcare funding proposals valued between $20M–$500M
- Partnered with senior leadership to develop an ethical AI governance framework addressing algorithmic safety and health equity
- Analyzed federal healthcare policies and produced executive summaries supporting institutional decision-making
- Coordinated operational workflows for School of Medicine and Advanced Medical Engineering Grand Rounds
- Contributed to interdisciplinary initiatives integrating healthcare technology, policy, and clinical innovation

Data Scientist (TuringMinds.ai), Bengaluru, India Sep 2022 – May 2023

- Performed exploratory analysis on datasets exceeding 2M records using Python and SQL, improving processing efficiency by 35%
- Implemented quality control procedures reducing analytical errors by 15% through validation and root cause analysis
- Troubleshot complex analytical pipelines using systematic debugging and verification methodologies
- Developed predictive models supporting data-driven decision making

PROJECTS

C.E.R.A. — Clinical Equipment Retrieval Assistant (Clinical AI System) Sep 2025 – Present

- Developed an enterprise-grade **Retrieval-Augmented Generation (RAG)** system to support biomedical engineers in troubleshooting ICU ventilators using manufacturer service manuals
- Designed hybrid semantic + keyword retrieval pipelines enabling precise identification of **error codes, part numbers, and repair procedures**, reducing manual documentation search time
- Implemented **medical device risk-management guardrails** aligned with FDA and **ISO 14971 principles**, including automated extraction and prioritization of safety warnings (WARNING/CAUTION/DANGER)
- Engineered automated generation of **maintenance work-order documentation** compatible with CMMS workflows, improving clinical engineering documentation efficiency

- Built structured comparison and checklist generation modules to convert technical manuals into **actionable diagnostic workflows** for hospital engineering environments

Emotion-Aware Tactile Feedback System for Deaf Users May 2025 – Dec 2025

- Designed a closed-loop wearable biofeedback system integrating **EEG biosignal acquisition with adaptive haptic feedback** to enhance emotional perception in live acoustic environments
- Developed real-time signal processing pipelines using EEG metrics (Frontal Alpha Asymmetry, Theta/Beta ratio) to dynamically modulate tactile stimulation
- Integrated Muse EEG headset data streams into custom software architecture enabling real-time physiological sensing and feedback control
- Implemented experimental evaluation workflows and analyzed user interaction data to assess system responsiveness and engagement outcomes
- Applied human-centered engineering principles to support assistive technology development for accessibility applications

CAPTAVATE — Wearable EEG Device Prototype Jan 2024 – May 2024

- Executed design verification testing on attention-detection algorithms by correlating predicted vs. observed outputs to validate neurofeedback accuracy
- Developed validation datasets and performed statistical analysis in MATLAB to evaluate real-time signal classification performance
- Authored technical documentation and validation reports following Good Documentation Practices (GDP) supporting device development lifecycle activities